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ADAMTS2 mediates epithelial–mesenchymal transition and inflammation in high-grade serous ovarian cancer: a study based on bioinformatic analyses and experiments

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Abstract

Background The importance of epithelial–mesenchymal transition (EMT) in tumour invasion and metastasis in high-grade serous ovarian cancer (HGSOC) has been highlighted in numerous studies, but genetic biomarkers for predicting EMT in HGSOC are still lacking.

Methods The role of EMT hallmarks and the relationship between EMT and the tumour microenvironment in HGSOC were examined based on transcriptomic data from 366 HGSOC patients in the TCGA dataset via the GSVA algorithm, the ESTIMATE method and Pearson correlation analyses. Furthermore, machine learning was applied to determine key EMT signatures and classify EMT subtypes. Moreover, the role of the key EMT signature ADAMTS2 in the behaviour and EMT process of HGSOC cells was detected via western blot, CCK8, transwell, wound healing and immunofluorescence assays.

Results The expression of EMT hallmarks (EMT score) was significantly associated with the progression, immune microenvironment and prognosis of HGSOC patients. Different machine learning algorithms identified *MMP2*, *ADAMTS2*, *FN1*, *THBS2*, *C3ORF80*, *FAP* and *POSTN* as key EMT-related signatures in HGSOC. Finally, qPCR, western blotting and IHC staining consistently revealed elevated expression of ADAMTS2 in five ovarian cancer tissues from HGSOC patients and five normal ovarian epithelial tissues from five uterine prolapse patients. Further in vitro experiments revealed that ADAMTS2 knockdown inhibited cell proliferation, migration, and invasion as well as the expression of TNF- α , IL-1 β and EMT markers (E-cadherin, N-cadherin, SLUG, and TWIST1), whereas ADAMTS2 overexpression promoted the above cellular behaviours and increased the expression of TNF- α , IL-1 β and EMT markers.

Conclusion Our study presents a new classifier to predict EMT and the immune microenvironment in HGSOC and identifies *ADAMTS2* as a novel regulator of the EMT process. These findings might promote the development of EMT-related immunotherapeutic strategies for HGSOC patients.

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Keywords Epithelial–mesenchymal transition, Machine learning, Tumour immune microenvironment, High-grade serous ovarian cancer, *ADAMTS2*

Background

Histologically, high-grade serous ovarian cancer (HGSOC) is a subtype of epithelial ovarian carcinoma (EOC) and is responsible for 70–80% of ovarian cancer-related deaths [1]. Owing to nonspecific and diverse symptoms, the early-stage diagnosis of HGSOC is difficult, thus leading to an unfavourable prognosis [2]. Patients diagnosed at an early stage have a 10-year survival rate of 55%, whereas the 10-year survival rate of patients diagnosed at an advanced stage is only 15% [3]. In recent years, great progress has been made in the treatment of HGSOC, such as the introduction of targeted therapies and antiangiogenic agents [4]. However, the clinical benefits are not ideal because of the occurrence of drug resistance [5] and relapse [6] and the complexity of the immunophenotype [7]. Thus, there is still much to learn about the molecular pathogenesis of HGSOC.

Epithelial–mesenchymal transition (EMT) refers to a process in which epithelial cells acquire the characteristics of mesenchymal cells [8]. Single-cell RNA sequencing has revealed that EMT-related genes are differentially expressed between HGSOC tumour cells and normal ovarian cells [9], and fibroblasts from metastatic HGSOC are more enriched in EMT progression than those from primary HGSOC [10], suggesting the importance of EMT in HGSOC tumorigenesis and progression. In addition, Wu et al. reported that *HMGA2* overexpression led to the transformation of ovarian surface epithelial cells in vitro and tumour formation in vivo, and further sequencing revealed that EMT-related genes and microRNAs may be responsible for the oncogenic role of *HMGA2* [11], again demonstrating the critical role of EMT in the initiation of HGSOC tumours and migration of HGSOC cells. These findings indicate that targeting EMT-related genes and modulating the EMT process may be promising strategies for the treatment of HGSOC patients. However, key EMT-related genes and EMT subtypes in HGSOC are not well characterized.

The tumour microenvironment is composed of stromal cells, the extracellular matrix, immune cells, blood vessels and other cell types [12]. It is closely associated with tumour growth, invasion, metastasis and therapeutic response [13]. The EMT process is directly regulated by the tumour microenvironment. For example, M2 tumour-associated macrophages in the tumour microenvironment produce TGF- β , thus promoting EMT in tumours [14]. A proteomic study in HGSOC by Leung et al. revealed that the EMT-related marker expression profile was altered in a carboplatin-resistant cell line and

was associated with the infiltration of multiple immune cells in the tumour microenvironment [15], indicating an interplay among EMT, the tumour microenvironment and drug sensitivity in HGSOC. However, whether EMT subtypes exhibit different immune microenvironments and therapeutic responses has not been elucidated in HGSOC. Answering the above question may provide guidance for the development of novel therapeutic strategies to improve the survival of HGSOC patients.

In the present study, the EMT score of HGSOC patients was calculated via RNA sequencing data. HGSOC patients were subsequently stratified into two EMT subtypes according to different clinical phenotypes, tumour mutations and functional annotations. Moreover, the tumour microenvironment, expression patterns of immune markers and sensitivity to paclitaxel were distinguished between the two EMT subtypes. In addition, four machine learning algorithms were used to identify key EMT signatures and construct an EMT subtype classifier. Further analysis and validation revealed that the classifier is capable of predicting the immune microenvironment status of HGSOC patients.

Materials and methods

Data source

The transcriptomes and survival information of 366 high-grade serous ovarian cancer (HGSOC) patients were downloaded from the UCSC Xena database (<https://xena.ucsc.edu>). The somatic mutation and copy number variation (CNV) data of 261 HGSOC patients were also sourced from the UCSC Xena database. In addition, the GSE32062 dataset, which includes gene expression data from 260 HGSOC patients, was used as an external validation set and was downloaded from the GEO database (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE32062>). A total of 200 EMT-related genes were derived from the “EPITHELIAL_MESENCHYMAL_TRANSITION” hallmark gene set in the MSigDB database (http://www.gsea-msigdb.org/gsea/msigdb/human/geneset/HALLMARK_EPITHELIAL_MESENCHYMAL_TRANSITION.html).

Evaluation of the tumour immune microenvironment

In the present study, the immune score, stromal score, ESTIMATE score and degree of immune cell infiltration were used to evaluate the tumour immune microenvironment in HGSOC patients. The ESTIMATE score, stromal score, and immune score of each HGSOC patient were computed using the estimate package (v1.0.13), following the algorithm proposed by Yoshihara et al. [16].

This method infers the fraction of stromal and immune cells in tumor samples based on gene expression data by applying single-sample Gene Set Enrichment Analysis (ssGSEA) to predefined gene signatures associated with stromal and immune components. Specifically, the stromal score reflects the presence of stromal cells, the immune score estimates the infiltration of immune cells, and the ESTIMATE score integrates both to infer tumor purity [16]. The infiltration of naive B cells, memory B cells, plasma cells, CD8 T cells, naive CD4 T cells, resting and activated memory CD4 T cells, follicular helper T cells, regulatory T cells, gamma delta T cells, resting and activated NK cells, monocytes, M0, M1 and M2 macrophages, resting and activated dendritic cells, resting and activated mast cells, eosinophils and neutrophils was determined via the CIBERSORT algorithm.

Differential expression analysis

The EMT score of each patient was calculated using the Gene Set Variation Analysis (GSVA) algorithm implemented in the *gsva* package (v1.42.0), based on the “HALLMARK_EPITHELIAL_MESENCHYMAL_TRANSITION” gene set from the MSigDB database. GSVA is a non-parametric, unsupervised method that estimates the variation of pathway activity over a sample population in an expression dataset. Unlike traditional enrichment methods that analyze groups of samples, GSVA transforms gene expression data into pathway-level scores for each individual sample [17], allowing us to quantitatively assess the EMT activity in each HGSOC patient. HGSOC patients were subsequently divided into low- and high-EMT score groups via the cut-off value determined using the *survminer* package (v0.4.9), which identifies the optimal threshold associated with survival outcomes. The differentially expressed genes (DEGs) between the low- and high-EMT score groups were identified via the *DESeq2* package, with adjusted *p* values < 0.05 and $|\log_2FC| > 2$ as cut-offs. GO and KEGG pathway enrichment analyses of the DEGs were performed via the *clusterProfiler* package.

Consensus clustering analysis

The *ConsensusClusterPlus* package was used to conduct consensus clustering to classify HGSOC patients into immune subtypes and EMT subtypes in the present study. To classify immune subtypes, the ssGSEA method was used to score the levels of 29 immune signatures (immune cells, pathways and functions), including aDCs, APC coinhibition, APC costimulation, B cells, CCR, CD8+ T cells, checkpoints, cytolytic activity, DCs, HLA, iDCs, inflammation-promoting factors, macrophages, mast cells, MHC class I, neutrophils, NK cells, parainflammation, pDCs, T-cell coinhibition, T-cell costimulation, T helper cells, Tfh cells, Th1 cells, Th2 cells, TIL,

Tregs, type I IFN response and type II IFN response. To classify the EMT subtypes, the expression patterns of the DEGs between the low- and high-EMT score groups were input to perform consensus clustering.

Candidate small molecules and immune/chemotherapeutic response

To screen candidate small molecules targeting EMT in HGSOC, we uploaded DEGs between the low- and high-EMT score groups to the CMap database (<https://clue.io/>). The expression levels of *CD80*, *CD86*, *CTLA4*, *PDCD1*, *PDL1* and *PDL2* were examined to evaluate the immunotherapeutic sensitivity of different EMT subtypes. In addition, paclitaxel/platinum-based chemotherapy is the standard first-line treatment for HGSOC [18]. Thus, the IC50 values of paclitaxel and cisplatin were predicted via the *pRRophetic* package to evaluate the chemotherapeutic response of HGSOC patients with different EMT subtypes.

Development and validation of the EMT classifier via machine learning

The EMT classifier was constructed based on the DEGs between high and low EMT score groups. To select robust EMT-related feature genes, we applied four machine learning algorithms: LASSO, SVM-RFE, XGBoost, and Random Forest. The intersection of the selected features from all four algorithms were used to build a logistic regression model as the EMT classifier. To train and validate the classifier, we randomly split the TCGA cohort (*n* = 366) into a training set (70%, *n* = 255) and a testing set (30%, *n* = 111). Feature selection and model construction were conducted using the training set, and model performance was evaluated on the testing set. The performance of the classifier was evaluated by ROC curve analysis and PCA. In addition, the accuracy of the classifier was further verified in the testing set and the external validation dataset.

Patients and tissue samples

Ovarian cancer tissues from five HGSOC patients who underwent primary debulking surgery and normal ovarian epithelial tissues from five uterine prolapse patients whose ovaries were removed during surgery were obtained from the Fourth Hospital of Hebei Medical University. After resection, all samples were immediately snap frozen in liquid nitrogen and stored at -80 °C until further use. The pathologic diagnosis of ovarian cancer tissues and normal ovarian epithelial tissues was confirmed by two senior pathologists. All experiments were conducted in accordance with the Declaration of Helsinki and approved by the Ethics Committee of the Fourth Hospital of Hebei Medical University (no. 2023KS157).

Informed consent was obtained from all individuals involved in the study.

Cell lines and cell culture

The noncancerous IOSE-80 ovarian epithelial cell line and two HGSOC cell lines, CAOV3 and OVCAR3, were obtained from the cell bank of the Chinese Academy of Sciences. IOSE-80 cells were cultured in IOSE-80 complete medium (Cas9X, Suzhou, China), and CAOV3 and OVCAR3 cells were cultured in RPMI 1640 (Thermo Fisher Scientific, CA, USA) supplemented with 10% foetal bovine serum (Thermo Fisher Scientific, CA, USA). All cells used in the present study were incubated at 37 °C in a humidified atmosphere of 5% CO₂.

qPCR analysis

Total RNA from the samples was extracted with TRIzol Reagent (Solarbio, Beijing, China). The quality of total RNA was determined with a NanoDrop 2000 (Thermo Fisher Scientific, USA). Next, total RNA was reverse transcribed into cDNA using Hifair® III 1st Strand cDNA Synthesis SuperMix for qPCR (Yeasen, Shanghai, China). The expression of key genes identified in our study was subsequently determined via a Q2000B real-time PCR detection system (LongGene, Hangzhou, China) with Hieff® qPCR SYBR Green Master Mix (Yeasen, Shanghai, China) at 95 °C for 3 min, 95 °C for 10 s, 60 °C, and 30 s for 40 cycles. The primers used in the present study are listed in Table 1. *GAPDH* was used as the reference gene, and the relative expression of key genes was calculated via the 2^{-ΔΔCT} method.

Table 1 Primers for qPCR

Gene name	Sequence (5' to 3')
MMP2	F: ATGATGCCTTTGCTCGTGC R: GAGTCCGCTCTTACCGTCAAA
ADAMTS2	F: CCAAGGCGACAAGTCAATATTC R: CAATGTCGTTGTGCTTCCCA
FN1	F: AGGCTTGAACCACTACGG R: GCCTAAGCACTGGCACAAACA
THBS2	F: AATAAACGAACCTCTCTTGTCTCT R: CGCACTTTGCTCTGCTAACCT
C3orf80	F: TCTCTGCGTTTCATTCGGTTC R: GAACCACAATACGAGGCAAGG
FAP	F: ACTTACCCTGCGTATGTAGGTCC R: ATGTCTGCCAGTCTTCCCTGA
POSTN	F: ATGGCACTCTGGGCATCGT R: ATGGCACTCTGGGCATCGT
GAPDH	F: GGAGTCCAATGGCGTCTTCA R: TGAGTCCTTCCACGATACCAAA

F Forward, R Reverse. *MMP2* Matrix metalloproteinase 2, *ADAMTS2* ADAM metalloproteinase with thrombospondin type 1 motif 2, *FN1* fibronectin 1, *THBS2* Thrombospondin 2, *C3orf80* Chromosome 3 open reading frame 80, *FAP* Fibroblast activation protein alpha, *POSTN* Periostin, *GAPDH* Glyceraldehyde-3-phosphate dehydrogenase

Western blotting

RIPA lysis buffer (Bozhen Biological, Shijiazhuang, China) was used to extract total protein from tissues and cells according to the manufacturer's instructions. Then, total protein was separated via SDS-PAGE and transferred to PVDF membranes, which were blocked with 5% BSA for 1 h at room temperature and then incubated with primary antibodies overnight at 4 °C and secondary HRP-conjugated goat anti-rabbit IgG (H+L) antibody (511203, 1:2000, ZENBIO, Chendu, China) for 1 h at room temperature. Thereafter, the blots were visualized using an ECL imaging system (JiaPeng, Shanghai, China). The primary antibodies used in the present study were against ADAMTS2 (A10272, 1:1000, ABclonal, Wuhan, China), β-actin (AC038, 1:2000, ABclonal, Wuhan, China), SLUG (WL01508, 1:1000, Wanleibio, Shenyang, China), TWIST1 (WL00997, 1:500, Wanleibio, Shenyang, China), N-cadherin (22018-1-AP, 1:3000, Proteintech, Wuhan, China), and E-cadherin (AF0131, 1:1000, Affinity, Jiangsu, China).

Immunofluorescence

Cells were seeded on coverslips and fixed in 4% paraformaldehyde. Then, the cells were permeabilized with 0.1% Triton X-100 and blocked with goat serum for 30 min. The cells were incubated with primary antibodies, including anti-IL-1β (16806-1-AP, 1:200, Proteintech, Wuhan, China) and anti-TNF-α (60291-1-Ig, 1:200, Wanleibio, Shenyang China), at 4 °C overnight. Afterwards, the cells were incubated with secondary antibodies, including goat anti-rabbit IgG-AF488 (4030-30, 1:200, SouthernBiotech, Birmingham, USA) and goat anti-mouse IgG, human ads-AF555 (1030-32, 1:200, SouthernBiotech, Birmingham, USA), and the nuclei were counterstained with DAPI. Images were captured with a fluorescence microscopic imaging system (Keyence, Osaka, Japan).

IHC staining

Slides of normal ovarian samples and HGSOC samples were first deparaffinized and rehydrated. The slides were subsequently incubated with epitope retrieval solution and peroxidase blocking solution for 10 min at room temperature. The slides were subsequently incubated with primary antibody (ADAMTS2, 1:40, Origene) for 1 h at room temperature and then incubated with a goat anti-mouse secondary antibody for 30 min at room temperature. After three washes in PBS buffer, the slides were stained with DAB and haematoxylin for visualization.

RNA interference, overexpression and cell transfection

si-*ADAMTS2* (5'-GGGAAUGACGUAAGAAAGATT-3') and si-NC (5'-UUCUCCGAACGUGUCACGUTT-3') were synthesized by Sangon Biotech (Shanghai, China). The CDS region of *ADAMTS2* was synthesized by

General Biol (Anhui, China) and ligated to the pcDNA3.1 vector to obtain the pcDNA3.1-*ADAMTS2* overexpression construct. Transfection was conducted using Lipofectamine 3000 (Thermo Fisher Scientific, MA, USA) according to the manufacturer's protocol. At 48 h post-transfection, the transfection efficiencies in the knock-down and overexpression groups were determined via western blotting.

Cell viability and colony formation assays

Cells were seeded at 2.5×10^4 /ml in a 96-well plate and cultured for 24 h. Then, the cells were transfected with si-NC, si-*ADAMTS2*, pcDNA3.1 empty vector or pcDNA3.1-*ADAMTS2* and cultured for another 48 h. Next, cell viability was assessed via the application of 10 μ l of CCK8 reagent (TargetMol, Shanghai, China). After 4 h of incubation, the optical density at 450 nm was measured.

Cells were seeded at a density of 500 per well in 24-well plates and were transfected with si-NC, si-*ADAMTS2*, pcDNA3.1 empty vector or pcDNA3.1-*ADAMTS2* and cultured for another 2–3 days. The supernatant was subsequently removed, and the cells were washed with PBS three times. Next, the cells were fixed with 4% paraformaldehyde for 10 min and stained with crystal violet for 1 min. Finally, the cells were washed with water to remove excess stain and air-dried for counting.

Wound healing, migration and invasion assays

Cells were seeded at 5×10^5 /well in 6-well plates and transfected with si-NC, si-*ADAMTS2*, pcDNA3.1 empty vector or pcDNA3.1-*ADAMTS2*. At 48 h posttransfection, the cells were scraped with a 200- μ l pipette tip to create a wound area. Then, PBS was applied gently to wash off the debris, and the cells were cultured in RPMI 1640 supplemented with 1% foetal bovine serum. Next, images of wound closure were obtained at 0 h and 24 h, respectively.

Transwell assays were carried out to evaluate cell migration and invasion. Two hundred microlitres of cells were resuspended at a density of 2×10^5 /ml and seeded in the upper chamber, and 600 μ l of culture medium supplemented with 10% foetal bovine serum was added to the lower chamber. Then, the cells were transfected with si-NC, si-*ADAMTS2*, pcDNA3.1 empty vector or pcDNA3.1-*ADAMTS2* and incubated in culture medium supplemented with 2% foetal bovine serum. At 48 h post-transfection, the cells in the upper chamber that did not migrate through the membrane were removed, and the cells that passed through the membrane into the lower chamber were fixed with 4% paraformaldehyde for 10 min. After several washes, the cells were stained with 0.1% crystal violet for 5 min, followed by microscopic observation and imaging.

Statistical analyses

R (version 4.2.2) was used to perform the statistical bioinformatic analyses. Survival-related EMT genes were screened via univariate Cox regression analysis. Kaplan–Meier (K–M) curves were plotted to measure the fraction of HGSOC patients living for a certain amount of time, and the log-rank test was used to evaluate the difference in survival between the two groups. The Wilcoxon or Kruskal–Wallis test was used to determine differences in data between two or three groups. Spearman correlation analysis was performed to evaluate the relationship between two variables. Fisher's exact test was used to compare the distributions of two categorical variables between two groups. For experimental assays, ImageJ (version 1.53) was first used to quantify the expression of proteins via western blot, immunofluorescence and IHC staining assays. GraphPad Prism (version 9.4.1) was used for the statistical analyses. Differences between two groups were evaluated by Student's t test, whereas differences among three groups were evaluated by one-way ANOVA. A p value < 0.05 was considered to indicate a statistically significant difference unless otherwise specified.

Results

Association of the EMT score with the clinical characteristics of HGSOC patients

After the EMT score of each patient was calculated via the GSVA algorithm, we used the EMT score to rank HGSOC patients from low to high and explored the relationship between the EMT score and clinical data (Fig. 1A). We observed that HGSOC patients with tumours and stage III–IV disease had significantly higher EMT scores than those without a tumour free and with stage I–II disease (Fig. 1B). In addition, we investigated the relationship between the EMT score and somatic mutations, which are commonly detected in HGSOC [19] (Fig. 1C). However, no significant difference in TMB was detected between the low- and high-EMT score groups (Fig. 1D). In addition, regarding the top five genes (*TP53*, *TTN*, *CSMD3*, *MUC16* and *FAT3*) with the highest mutation frequency in HGSOC, there was no significant difference in the EMT score between the wild-type and mutated types (Fig. 1E–I).

Close correlation of the EMT score and tumour immune microenvironment

We applied the ssGSEA algorithm to calculate the abundance of 29 immune hallmarks to evaluate the landscape of immune infiltration in HGSOC. HGSOC patients were subsequently divided into three immune clusters by the consensus clustering method, which were defined as the low-, medium- and high-immunity groups (Fig. 2A). To further confirm the immune status of the three groups,

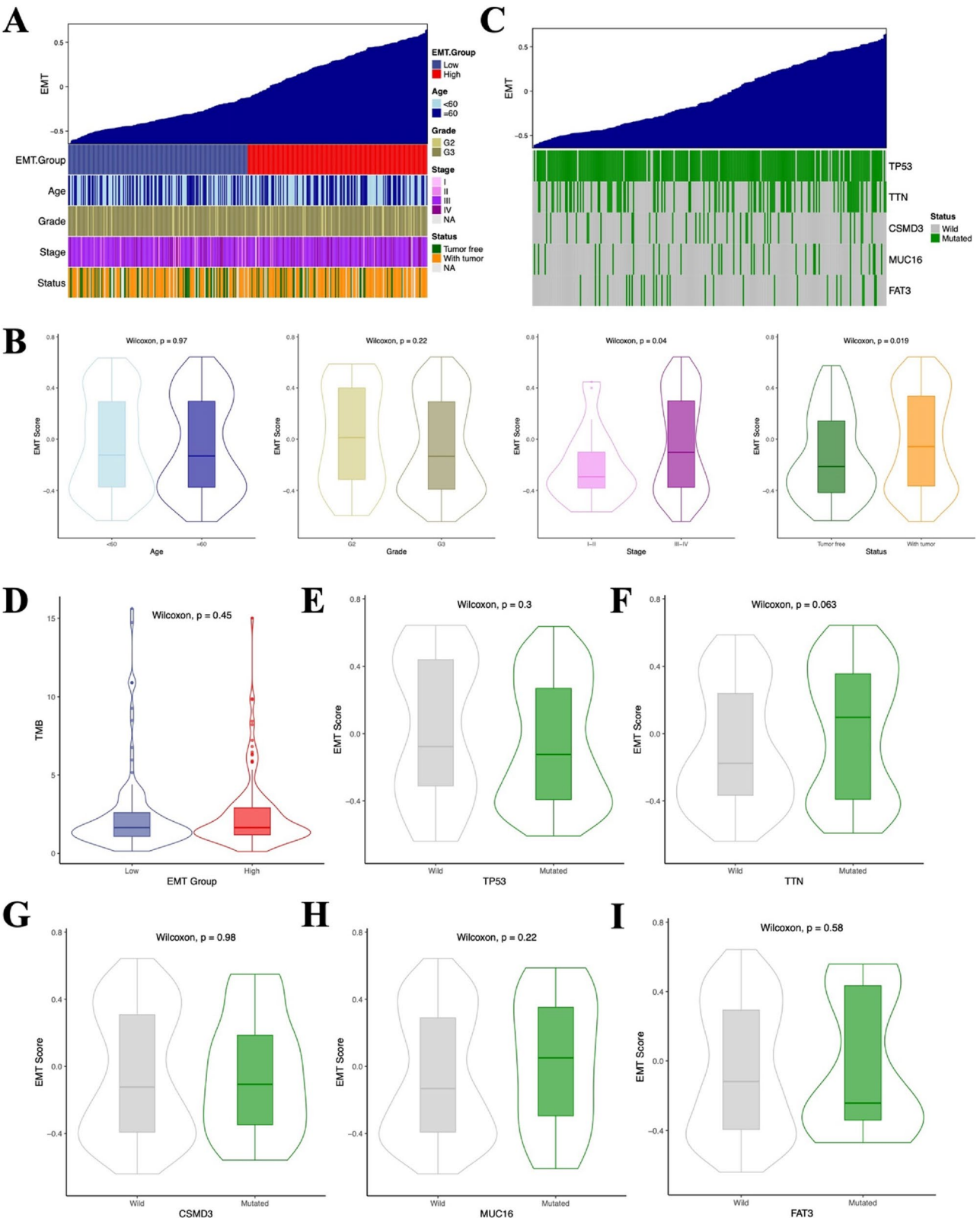


Fig. 1 Relationships between the EMT score and clinical characteristics in HGSC patients. **A** Global view of the relationship between the EMT score and the clinical phenotypes of HGSC patients. **B** Violin plots of the EMT score stratified by age, grade, stage and status. **C** Global view of the relationship between the EMT score and somatic mutations in HGSC patients. **D** Violin plot of TMB stratified by the EMT score. **E-I** Violin plots of the EMT score stratified by the mutational status of the top five genes with the highest mutation frequency

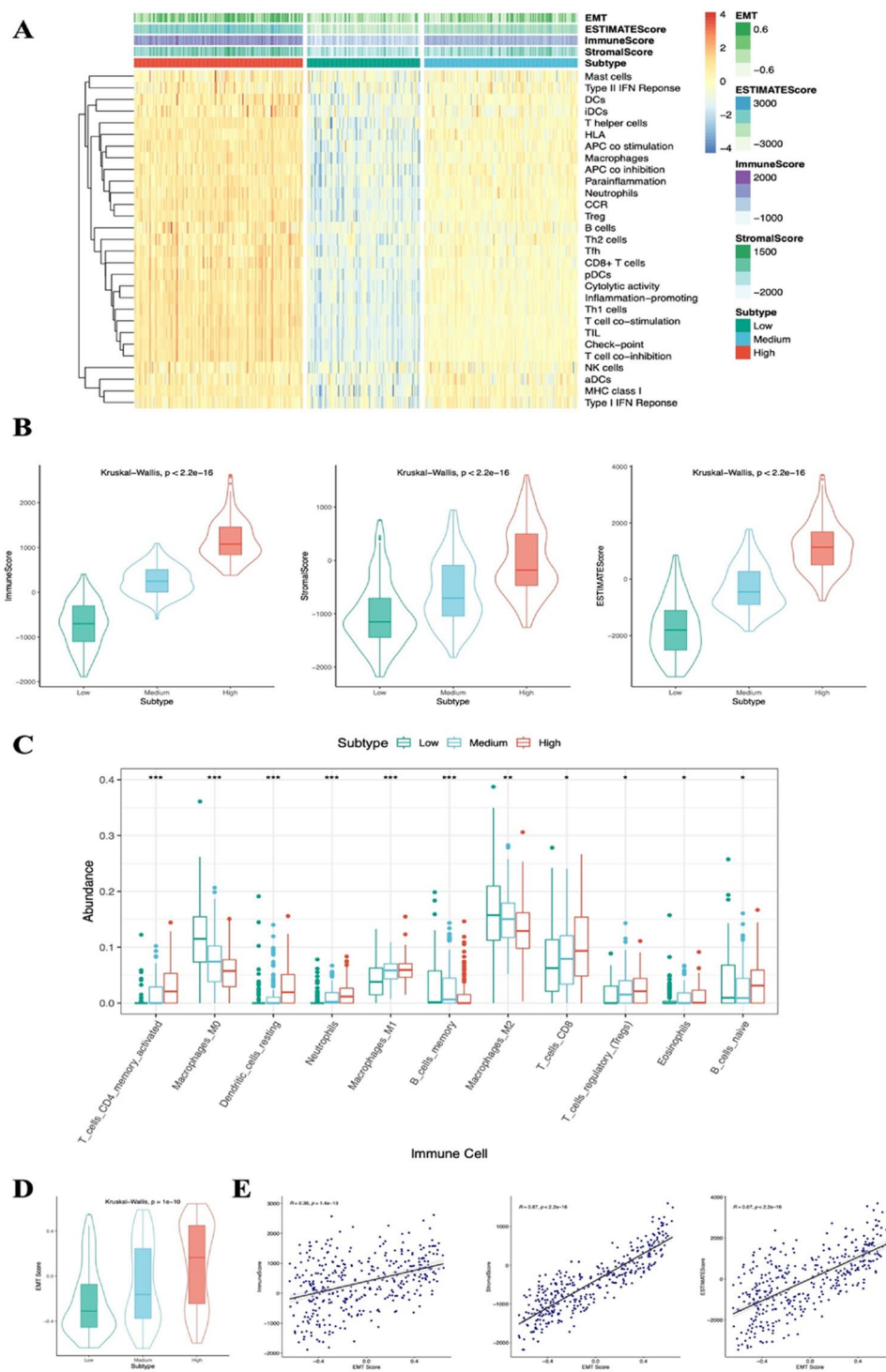


Fig. 2 Relationships between the EMT score and the tumour immune microenvironment in HGSC. **A** HGSC patients were clustered into three immune subgroups based on immune hallmarks. **B** Comparison of the immune score, stromal score and ESTIMATE score among the three immune subgroups. **C** Comparison of immune cells among the three immune subgroups. **D** Comparison of the EMT scores among the three immune subgroups. **E** Correlations between the EMT score and the immune/stromal/ESTIMATE scores. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.0001$

we performed ESTIMATE and CIBERSORT analyses. We observed that the high-immunity subgroup had the highest immune score, stromal score and ESTIMATE score, followed by the medium- and low-immunity

subgroups (Fig. 2B). Additionally, the enrichment of immune cells was significantly different among the three subgroups. For example, immunosuppressive M2 macrophages showed the highest abundance in the

low-immunity group, and the enrichment of CD8+T cells, the end effectors of cancer immunity, was highest in the high-immunity group (Fig. 2C). We subsequently explored the relationship between the EMT score and tumour immune microenvironment. We observed that the high-immunity group had the highest EMT score (Fig. 2D) and that the EMT score was significantly positively correlated with the immune score ($\text{cor}=0.38$), stromal score ($\text{cor}=0.87$) and ESTIMATE score ($\text{cor}=0.67$) (Fig. 2E), indicating a close relationship between the degree of EMT and immune status in HGSOc patients.

Identification of two EMT subtypes in HGSOc

Considering the above findings, we then divided HGSOc patients into low- and high-EMT score groups. We found that patients in the low-EMT score group had better survival than those in the high-EMT score group (Fig. 3A). To further investigate the molecular mechanism of EMT in HGSOc tumorigenesis, we sought to construct a novel EMT-based classification system for HGSOc patients to better characterize the role of EMT in HGSOc. First, differential expression analysis was performed between the low- and high-EMT score groups, and a total of 343 DEGs, including 141 upregulated and 202 downregulated genes in the high-EMT score group, were identified (Fig. 3B). Functional analyses revealed that these DEGs were enriched mainly in extracellular matrix organization, the collagen metabolic process, protein digestion and absorption, the PI3K-Akt signalling pathway, proteoglycans in cancer, and ECM-receptor interactions, which are related to the tumour microenvironment and tumorigenesis (Fig. 3C and D). In addition, the waterfall plot revealed that the most common mutation among the DEGs was missense mutation, and the top 20 mutated DEGs were upregulated in the high-EMT score group, with the exceptions of *COL2A1*, *MAP3K19*, *PADI3*, *DMBT1* and *MAGEA11* (Fig. 3E). Next, based on the expressions of DEGs, consensus clustering was applied to HGSOc patients, and two EMT subtypes (EMT subtype 1 and EMT subtype 2) were identified. As shown in Fig. 3F, subtype 1 patients had significantly higher EMT scores than subtype 2 patients. Moreover, Kaplan–Meier curves revealed that EMT subtype 1 patients tended to have worse survival than EMT subtype 2 patients ($p=0.052$, Fig. 3G).

Different clinical phenotypes, mutation patterns and functional annotations between EMT subtypes

Thereafter, we explored the characteristics of the two EMT subtypes. First, we compared the clinical phenotypes between the two EMT subtypes. We found that EMT subtype 2 patients were more likely to be tumour free and have stage I-II disease (Fig. 4A). GSVA was then performed to detect the potential molecular mechanisms

that may determine EMT subtypes in HGSOc. Figure 4B shows a heatmap of the top 10 pathways enriched in the two EMT subtypes. Specifically, pathways related to DNA repair and gene expression, such as the spliceosome, base excision repair, RNA degradation, nucleotide excision repair, RNA polymerase, and mismatch repair, were enriched in EMT subtype 2, whereas ECM-related pathways, such as ECM receptor interactions, focal adhesion, leukocyte transendothelial migration, and regulation of the actin cytoskeleton, were enriched in EMT subtype 1.

In addition, genomic alterations contribute to the occurrence of EMT. Thus, we analysed CNVs and somatic mutations in the two EMT subtypes. Interestingly, we observed that EMT subtype 2 had a significantly greater frequency of copy number amplifications and deletions (Fig. 4C). Although no significant difference in TMB was detected between the two EMT subtypes (Fig. 4D), some differences in somatic mutations were detected between the two EMT subtypes. Specifically, the common top three mutated genes in the two subtypes were *TP53*, *TTN* and *CSMD3*, followed by *DNAH3* and *MUC16* in subtype 1 and *MUC16* and *FAT3* in subtype 2. In addition, the mutational frequency of *TNN* in subtype 1 (47.31%) was significantly greater than that in subtype 2 (31.55%) (Fig. 4D).

Differences in the tumour immune microenvironment and response to targeted therapy between EMT subtypes

Considering that the EMT score is closely associated with the tumour microenvironment, we examined whether the two EMT subtypes possessed different tumour microenvironments. We observed that EMT subtype 1 patients had significantly higher immune scores, stromal scores and ESTIMATE scores (Fig. 5A). In addition, the CIBERSORT results also revealed significant differences in the infiltration of multiple immune cells between the two subtypes (Fig. 5B). Moreover, we found that most patients (55.81%) in EMT subtype 1 fell into the high-immunity subgroup, whereas more patients in EMT subtype 2 than subtype 1 (10.85%) had low-immunity HGSOc (34.18%) (Fig. 5C). In addition, we detected the expression of immune checkpoint proteins in the two subtypes. As shown in Fig. 5D, the abundance of *CD80*, *CD86*, *CTLA4*, *PDCD1* and *PDL2* was significantly lower in EMT subtype 2 patients, indicating that patients in EMT subtype 2 may have a worse response to immunotherapeutic treatment.

Next, we sought to investigate whether the two EMT subtypes also had different sensitivities to clinical drugs. The chemotherapy used for HGSOc includes paclitaxel and cisplatin [20]. As shown in Fig. 5E, the IC₅₀ value of paclitaxel in subtype 1 was significantly lower than that in subtype 2, indicating that HGSOc patients in subtype 1 were more sensitive to paclitaxel, but no significant

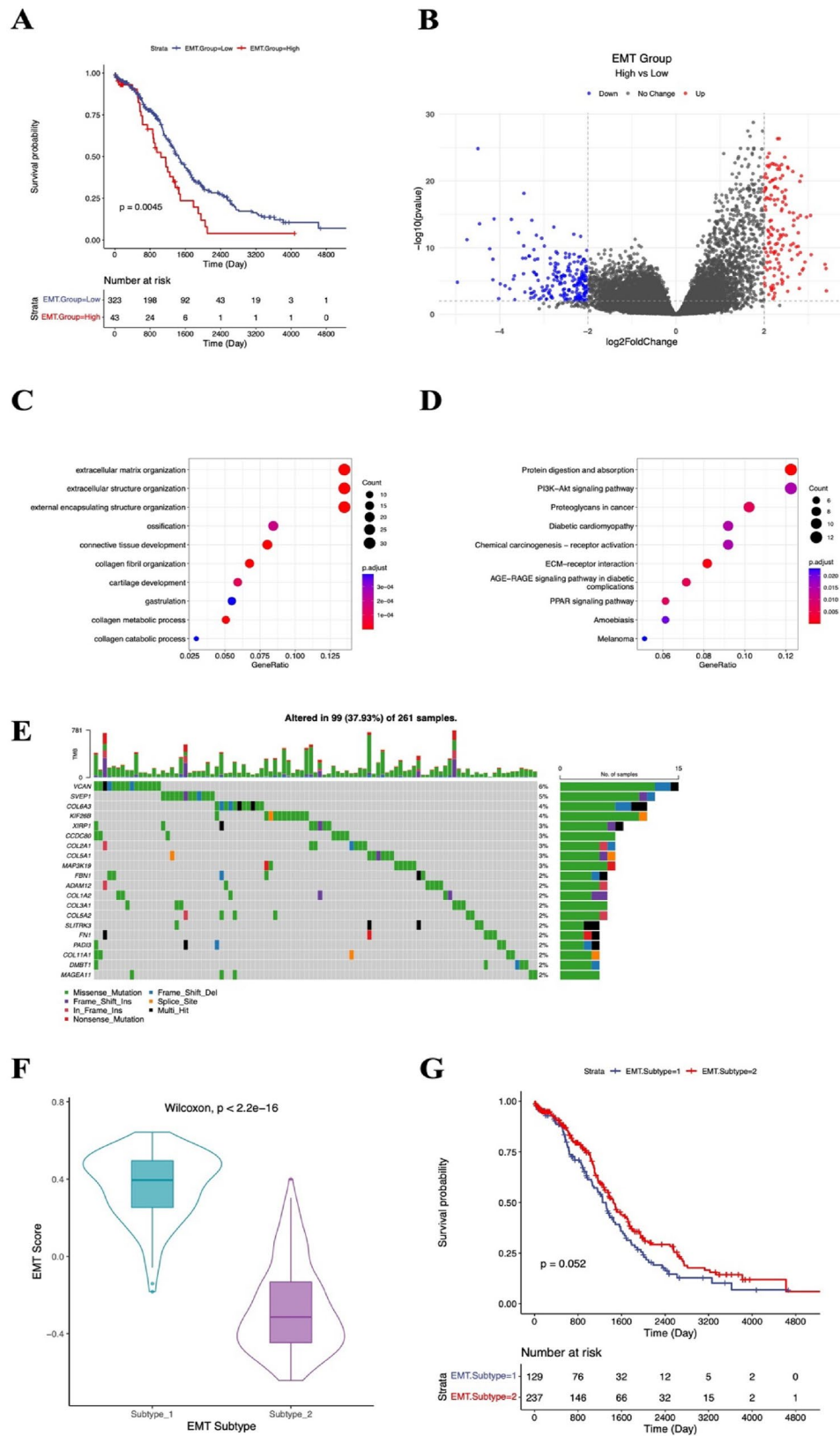


Fig. 3 Identification of two EMT subtypes in HGSOC. **A**, Kaplan–Meier curves showing significantly better survival in the low-EMT score group. **B** Volcano plot showing the differentially expressed genes (DEGs) between the low- and high-EMT score groups. **C** GO enrichment analysis of DEGs. **D** KEGG pathway enrichment analysis of DEGs. **E** Waterfall plot showing the mutation frequency of the top 20 frequently mutated DEGs. **F** Comparison of EMT scores between subtypes 1 and 2. **G** Kaplan–Meier curves showing the survival status of patients in subtypes 1 and 2

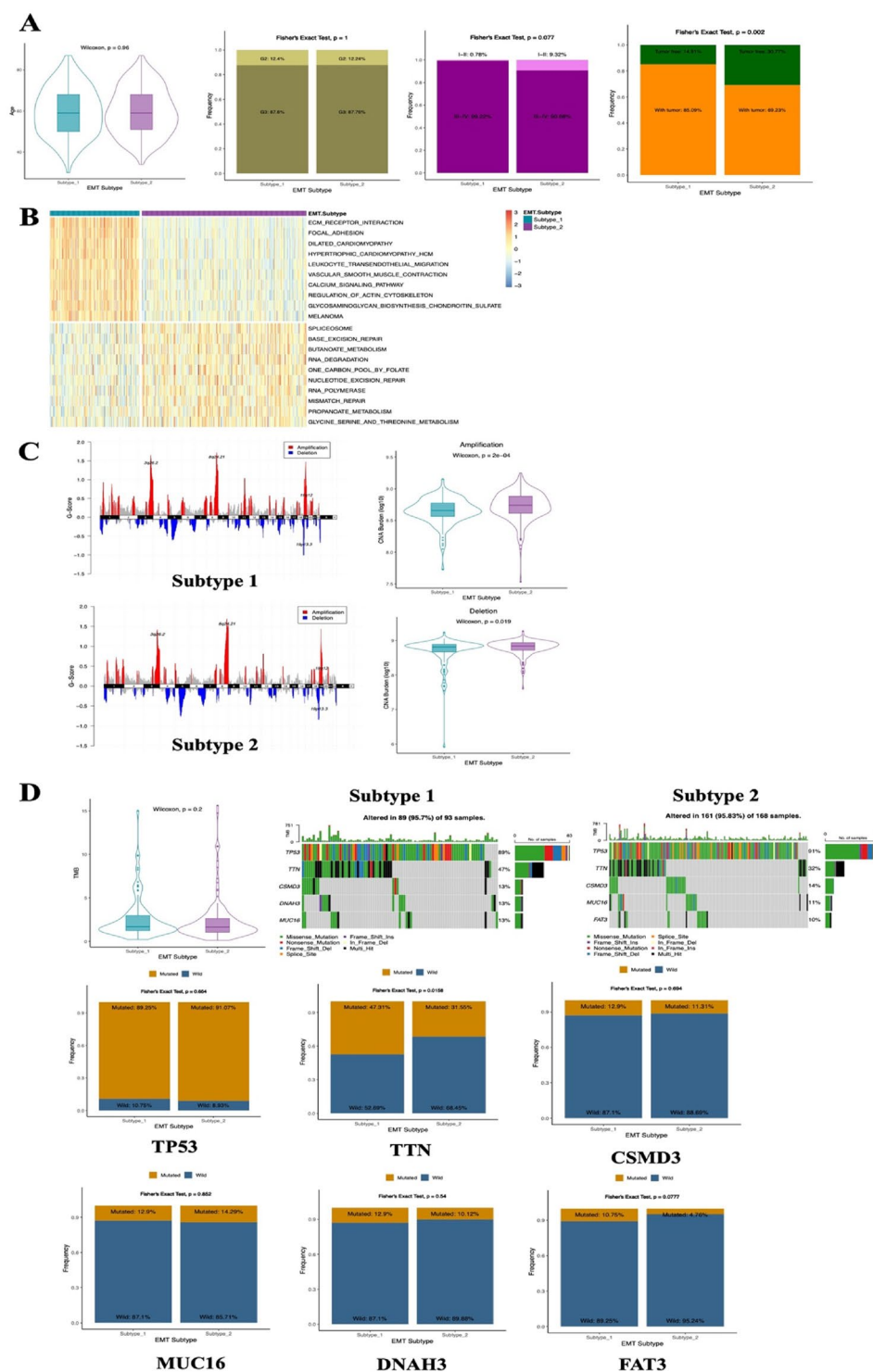


Fig. 4 Different clinical phenotypes, functional annotations and genomic alterations between the two EMT subtypes. **A** Comparison of age, grade, stage, and tumour status between the two EMT subtypes. **B** GSEA of enriched pathways between the two EMT subtypes. **C** Frequency of CNVs between the two EMT subtypes. **D** Frequency of somatic mutations between the two EMT subtypes

difference in the IC50 value of cisplatin was detected between the two subtypes. Moreover, we tried to identify novel potential drugs targeting DEGs to classify EMT subtypes via the CMap database, and 45 chemicals

enriched in 35 modes of action mechanisms were identified (Fig. 5F). These drugs and their corresponding mechanisms of action might improve immunotherapies that target EMT characteristics in HGSC.

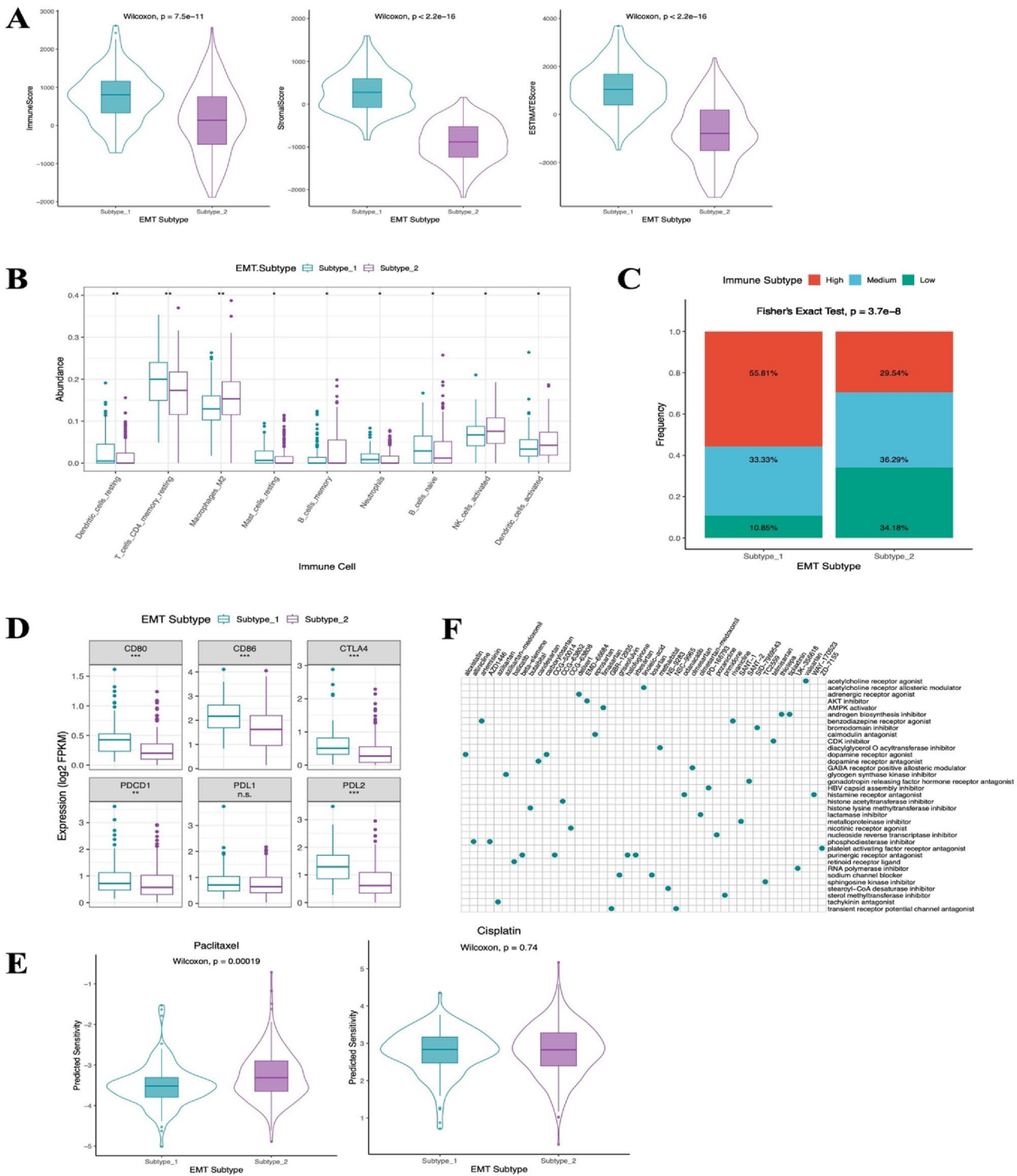


Fig. 5 Different tumour microenvironments and therapeutic sensitivities between the two EMT subtypes. **A** Comparison of immune/stromal/ESTIMATE scores between the two subtypes. **B** Comparison of immune cell infiltration between the two subtypes. **C** Distribution of immunity subgroups in the two subtypes. **D** Comparison of immune checkpoint molecule expression between the two subgroups. **E** Comparison of the IC50 values of paclitaxel and cisplatin between the two subtypes. **F** Identification of novel compounds and related modes of action targeting DEGs via the CMap database. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.0001$, ^{ns} $p > 0.05$

Construction and validation of a classifier distinguishing two EMT subtypes

The EMT classifier was constructed based on the 343 DEGs between high and low EMT score groups. To select robust EMT-related feature genes, we applied four machine learning algorithms: LASSO, SVM-RFE, XGBoost, and Random Forest. Among these algorithms, LASSO, SVM-RFE and XGBoost identified 29, 33 and 28 feature genes, respectively. As for random forest, it yielded 151 genes under the optimal number of decision trees, from which the top 30 genes with the highest feature importance were selected. The intersection of the selected features from all four algorithms resulted in seven common genes, including *MMP2*, *ADAMTS2*, *FN1*, *THBS2*, *C3ORF80*, *FAP* and *POSTN* (Fig. 6A). To further demonstrate the importance of key features, we performed PCA on the mRNA expression profiles of all genes, EMT-related genes associated with survival, EMT-related DEGs and key features and found that EMT subtypes 1 and 2 were distinctly distributed based on the expression of key features (Fig. 6B). Next, these seven genes were used to build a logistic regression model as the EMT classifier. To evaluate the performance of the classifier, we utilized ROC curves, which confirmed the accuracy and reliability of the classifier, with an AUC of 0.999 in the training set and an AUC of 0.996 in the testing set (Fig. 6C). Moreover, an external dataset was used to further validate the reliability of the classifier. The HGSOC patients in the validation dataset were divided into two EMT subtypes, which had a significant effect on the tumour microenvironment, as shown in Fig. 6D. Finally, we detected the expression of key features in HGSOC and adjacent control samples and observed that the expression of *ADAMTS2* was significantly increased, whereas the expression of *C3ORF80*, *FAP*, *FN1*, *POSTN* and *THBS2* was significantly decreased in tumour tissues compared with adjacent normal tissues (Fig. 6E).

Upregulation of ADAMTS2 both in vivo and in vitro

Considering the dual role of *ADAMTS2* in both the ECM and immune response, we focused on the role of *ADAMTS2* in HGSOC. Thereafter, we first detected its protein expression levels in HGSOC tissues and cell lines. The results of the western blot revealed that the expression of *ADAMTS2* was significantly greater in cancer tissues than in normal ovarian tissues (Fig. 7A). Meanwhile, immunohistochemical staining was also performed to evaluate *ADAMTS2* protein expression in clinical tissue specimens. In normal ovarian tissues, *ADAMTS2* staining was minimal or absent. In contrast, HGSOC tissues exhibited diffuse and strong cytoplasmic staining of *ADAMTS2* across the tumour regions. To provide a quantitative assessment, the percentage of positively stained area (%Area) was calculated. The results showed

a significantly higher *ADAMTS2*-positive area in cancer tissues compared with normal tissues (Fig. 7B). These results were consistent with the qPCR results. To further investigate the role of *ADAMTS2* in HGSOC, we detected whether two HGSOC cell lines (CAOV3 and OVCAR3) could be used for the following *ADAMTS2* functional assays. The western blot results revealed that *ADAMTS2* expression was much greater in the two HGSOC cell lines than in the normal ovarian surface epithelial cell line (IOSE-80) (Fig. 7C), which was consistent with the expression pattern we observed in vivo.

Promotion of HGSOC cell proliferation by ADAMTS2

To explore the effect of *ADAMTS2* on HGSOC cell behaviour, *ADAMTS2* knockdown and overexpression experiments were conducted. Western blot analysis revealed that both *ADAMTS2* knockdown and overexpression were effective in the two HGSOC cell lines (Fig. 8A). We observed that the viability of HGSOC cells was strongly inhibited by *ADAMTS2* knockdown and enhanced by *ADAMTS2* overexpression (Fig. 8B). In addition, colony formation assays revealed that *ADAMTS2* knockdown suppressed the number of cell colonies, whereas *ADAMTS2* overexpression increased the capacity for colony formation in both HGSOC cell lines (Fig. 8C). These results suggest that *ADAMTS2* could promote cell proliferation in HGSOC, acting as an oncogene.

Promotion of HGSOC cell invasion and migration by ADAMTS2

Next, we investigated whether *ADAMTS2* also affects the migration and invasion of HGSOC cells. Transwell assays revealed less invasion into lower chambers in the si-*ADAMTS2* group than in the si-NC group, while overexpression of *ADAMTS2* increased the ability of HGSOC cells to invade into the lower chamber (Fig. 9A and B). Moreover, the results of the wound healing assays revealed that knockdown of *ADAMTS2* inhibited cell migration, while the wound healing rate in the *ADAMTS2* overexpression group was approximately 2-fold greater than that in the control group (Fig. 9C and D). These results suggest that *ADAMTS2* promotes the migration and invasion of HGSOC cells.

Involvement of ADAMTS2 in the EMT process and tumour microenvironment in HGSOC

In the present study, *ADAMTS2* was retrieved from bioinformatic analyses of the EMT process, which is crucial in tumour metastasis. Thus, we then explored whether alteration of *ADAMTS2* expression would influence the EMT process by detecting the expression of EMT-related markers. We found that knockdown of *ADAMTS2* increased the expression of the epithelial cell marker

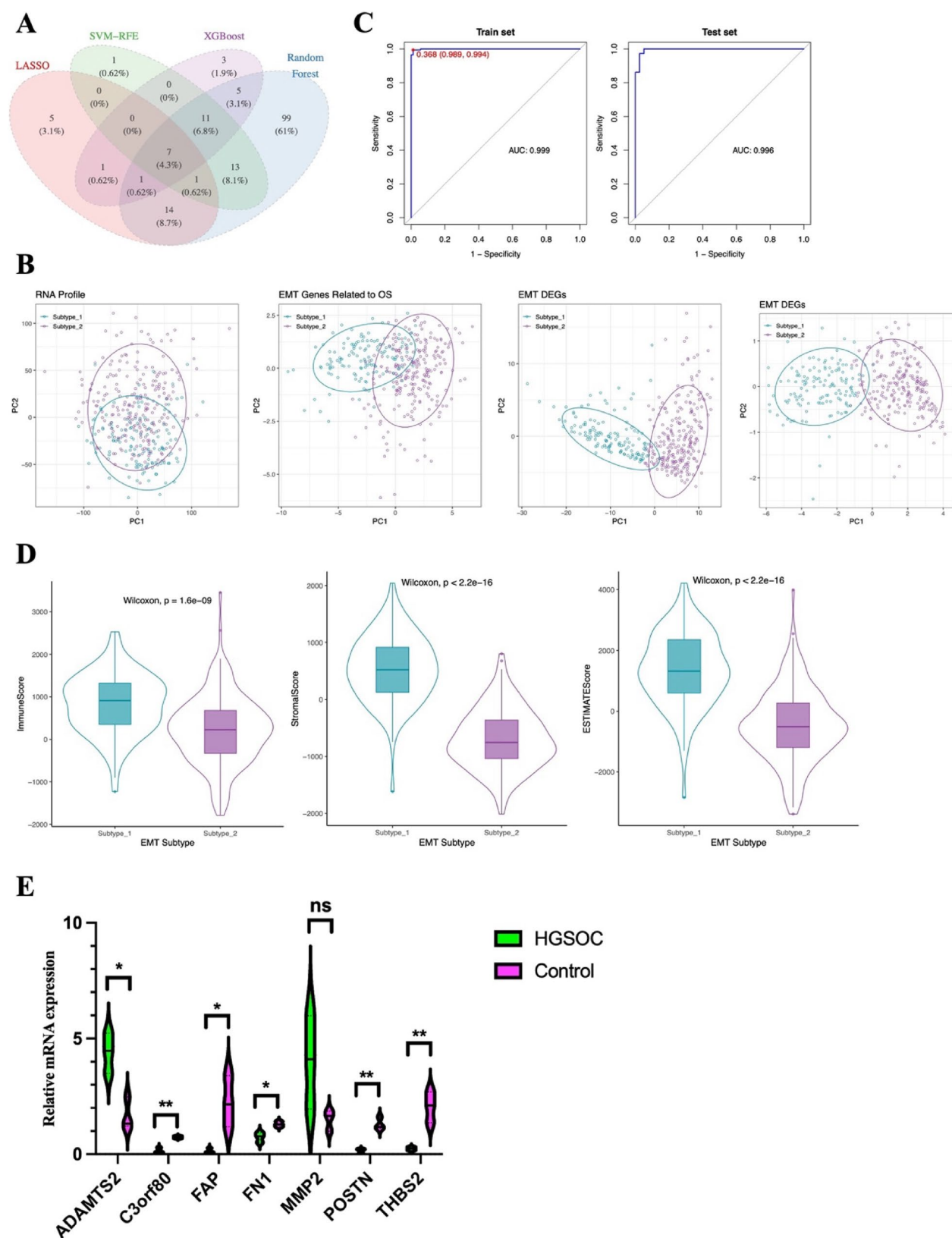


Fig. 6 Construction and validation of the classifier for predicting the EMT subtype. **A** Venn diagram showing seven common key features identified by the four machine learning methods. **B** PCA based on expressions of all genes, EMT genes, EMT-related DEGs and key features. **C** ROC curves of the training and testing sets. **D** Immune/stromal/ESTIMATE scores to validate the classifier in GSE32062. **E** Comparison of the expression of key features between tumour and adjacent normal tissues. * $p < 0.05$, ** $p < 0.01$, nsp > 0.05

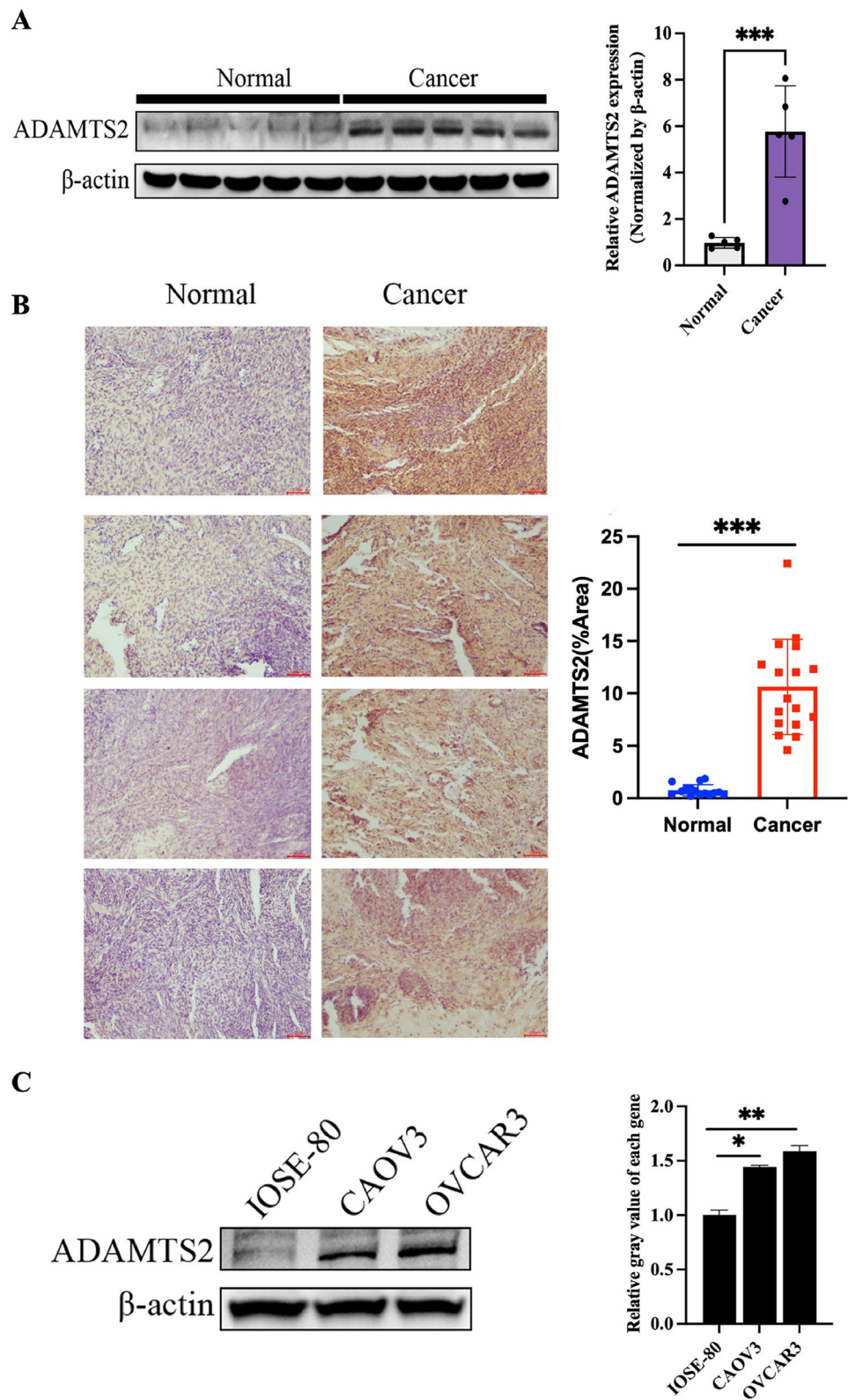


Fig. 7 Expression of ADAMTS2 in HGSOC tissues and cell lines. **A** Protein expression level of ADAMTS2 detected in normal ovarian tissues and cancer tissues via western blotting. The first five lanes correspond to normal samples, and the last five lanes correspond to cancer samples. **B** Representative images of IHC staining used to show diffuse cytoplasmic expression of ADAMTS2 in normal ovarian tissues (left) and cancer tissues (right) with quantitative analysis of the percentage of positively stained area. **C** Representative images of western blots showing the expression levels of ADAMTS2 in normal ovarian surface epithelial cell lines (IOSE-80) and two HGSOC cell lines (CAOV3 and OVCAR3). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

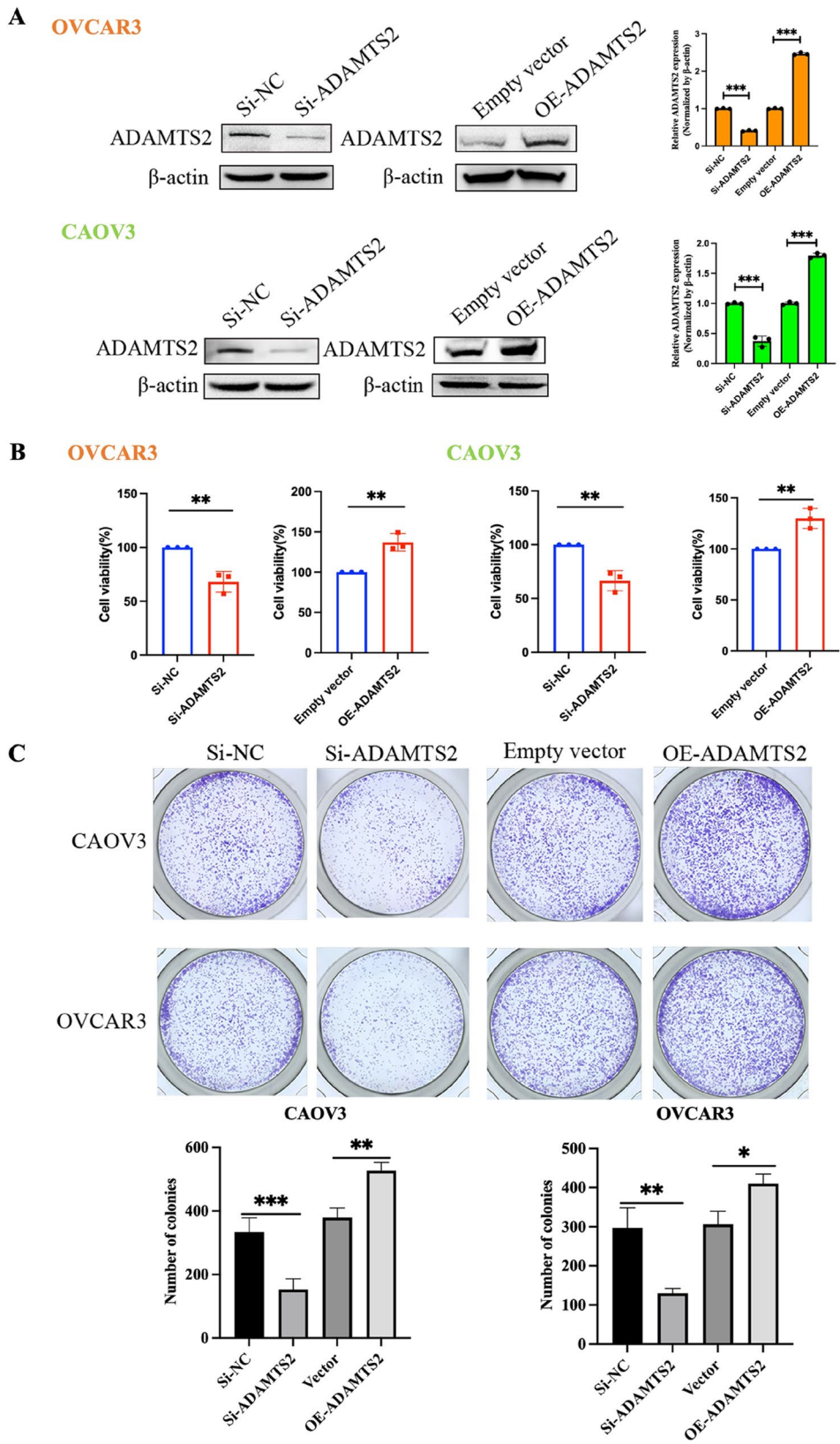


Fig. 8 Effects of ADAMTS2 knockdown or overexpression on HGSOC cell proliferation. **A** Transfection efficiency of ADAMTS2 knockdown and overexpression determined via western blotting. **B** Comparison of cell viability determined by a CCK9 assay between the knockdown/overexpression and control groups. **C** Representative images of colony formation assays in the knockdown/overexpression and control groups. All experiments were performed in two HGSOC cell lines (OVCA3 and CAOV3) and were repeated three times. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

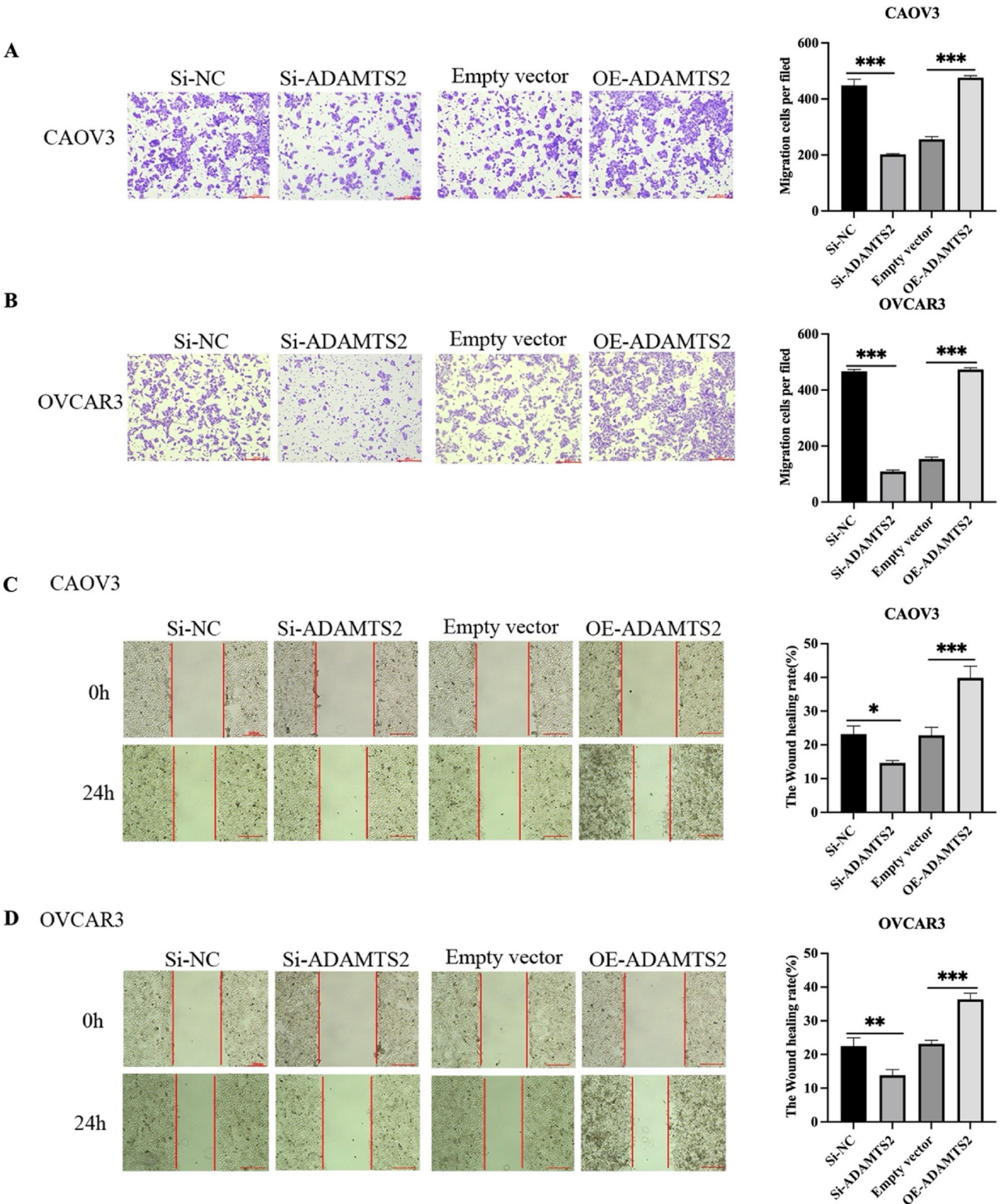


Fig. 9 Effects of ADAMTS2 knockdown or overexpression on the migration and invasion of HGSOc cells. Transwell invasion assays of CAOV3 (**A**) and OVCAR3 (**B**) cells to detect the invasion capacity of cells in the knockdown/overexpression and control groups. Wound healing assays of CAOV3 (**C**) and OVCAR3 (**D**) cells to compare the migration capacity between the knockdown/overexpression and control groups. All experiments were performed in triplicate. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

E-cadherin and reduced the expression of the mesenchymal cell marker N-cadherin and the EMT transcription factors SLUG and TWIST1 (Fig. 10). Conversely, the expression of N-cadherin, SLUG and TWIST1 was elevated, whereas the expression of E-cadherin was downregulated when ADAMTS2 was overexpressed (Fig. 10). In addition, via bioinformatic analyses, we revealed a close relationship between EMT and the tumour immune microenvironment. Thus, we also detected TNF- α and IL-1 β expression when ADAMTS2 was down- and upregulated. Downregulation of ADAMTS2 decreased TNF- α and IL-1 β expression, whereas upregulation of ADAMTS2 increased their expression in HGSOC cells (Figs. 11 and 12).

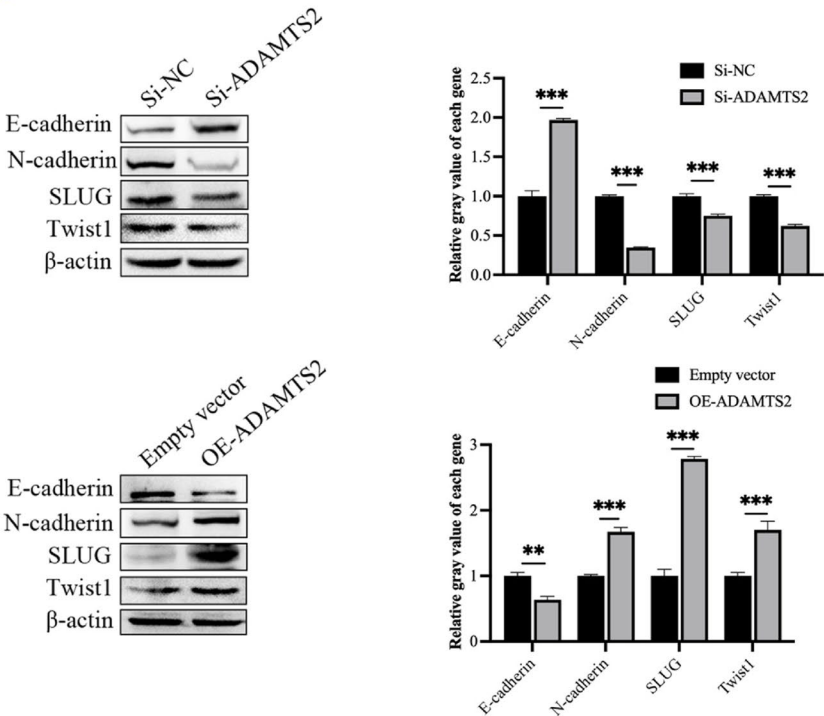
Discussion

The progress of immunotherapeutic treatment for HGSOC has not been promising because of the heterogeneity of the tumour microenvironment, which leads to differences in immune status and immunophenotype [7, 21, 22]. Thus, methods to distinguish different tumour microenvironments may improve the immunotherapeutic response of HGSOC patients. Many studies have shown crosstalk between EMT and the tumour microenvironment in several cancer types. For example, an elevated population of CD4⁺FOXP3⁺Tregs and increased expression of immune checkpoint proteins were found to be closely associated with the EMT status in non-small cell lung cancer [23]. Guo et al. reported disturbance of the EMT network and heavy infiltration of immune cells in sarcomatoid bladder cancer [24]. Acetylation of the EMT transcription factor Snail induces the expression of *CCL2* and *CCL5*, thus promoting the recruitment of macrophages into the tumour microenvironment [25]. In our study, we also found a close relationship between EMT and the tumour microenvironment. In the EMT subtype model, patients in subtype 1 with higher EMT scores had higher immune/stromal/ESTIMATE scores and a greater abundance of immune checkpoint molecules, such as *CTLA4*, *CD80* and *PDCD1*, suggesting that more research is needed to explore downstream molecular mechanisms involved in the crosstalk between EMT and the tumour microenvironment in HGSOC to benefit clinical application. Additionally, the EMT subtype classifier we developed may have the potential to predict the tumour immune status and immunotherapeutic response of HGSOC patients.

In the present study, we applied four machine learning methods and extracted the overlapping signatures as key EMT-related signatures in HGSOC to build a classifier. Each machine learning method has its own advantages and disadvantages [26, 27]. Therefore, in the present study, we adopted an integrated strategy in which the overlapping signatures identified by the four machine

learning methods were extracted as key EMT-related signatures to build the following classifier. These genes included *MMP2*, *ADAMTS2*, *FN1*, *THBS2*, *C3ORF80*, *FAP* and *POSTN*. *MMP2*, a matrix metalloproteinase, plays an important role in the metastasis of ovarian cancer, as its knockdown inhibits the proliferation, invasion and migration of ovarian cancer cells [28]. In addition, Zeng reported that the protein level of *MMP2* was increased in ovarian cancer tissues compared with normal tissues and that its mRNA expression level was negatively associated with the overall survival of ovarian cancer patients [29]. During EMT in renal tubular cells, active *MMP2* expression is required [30]. In human malignant glioma cells, chemokine receptor 7 promotes EMT by enhancing the expression of *MMP2/9* [31]. However, its role in mediating EMT in ovarian cancer has not been fully elucidated. *ADAMTS2* is important for the organization of fibrillar collagen. Mesenchymal-origin cells and fibroblasts are the main cells expressing *ADAMTS2*, and in vitro, its expression can be increased by TGF- β [32]. In gastric cancer, the expression of *ADAMTS2* is elevated in tumour tissues and is associated with overall survival. However, its function in EMT and ovarian cancer has not been well studied. *FN1* is also an ECM glycoprotein that regulates cell migration, differentiation and adhesion [33, 34]. In addition, it has been reported to play a key role in tumorigenesis and metastasis in gastric cancer [35], head and neck squamous cell carcinoma [36] and clear cell renal cell carcinoma [37]. A recent study in serous ovarian cancer revealed that the lncRNA *PTAL* interacted with miR-101 to increase the expression of *FN1*, thus promoting the metastasis of ovarian cancer cells by regulating EMT [38]. *THBS2* is involved in regulating cell adhesion, apoptosis, angiogenesis and motility by interacting with other proteins [39]. In gastric cancer, colorectal cancer and breast cancer, *THBS2* plays a key role in migration, invasion, the tumour microenvironment and EMT [40–42]. In ovarian cancer, the expression of *THBS2* was found to be significantly correlated with microvessel density [43]. Ramus and Gorringer reported that the expression of *THBS2* was associated with survival and infiltrative invasion in patients with mucinous ovarian carcinoma [44]. However, its exact role in EMT and the tumour microenvironment of HGSOC remains unclear. The expression of *FAP* has been found to be elevated in many cancer types, such as colorectal cancer, gastric cancer, brain cancer, pancreatic cancer, breast cancer, myeloma, and melanoma. Studies have shown that *FAP* regulates tumour cell behaviour, EMT, immunosuppression, stem cells and drug resistance through multiple mechanisms [45]. Specifically, in ovarian cancer, the knockdown of *FAP* decreases cell proliferation, tumour growth and volume, whereas the overexpression of *FAP* increases cell proliferation, invasion

A CAOV3



B OVCAR3

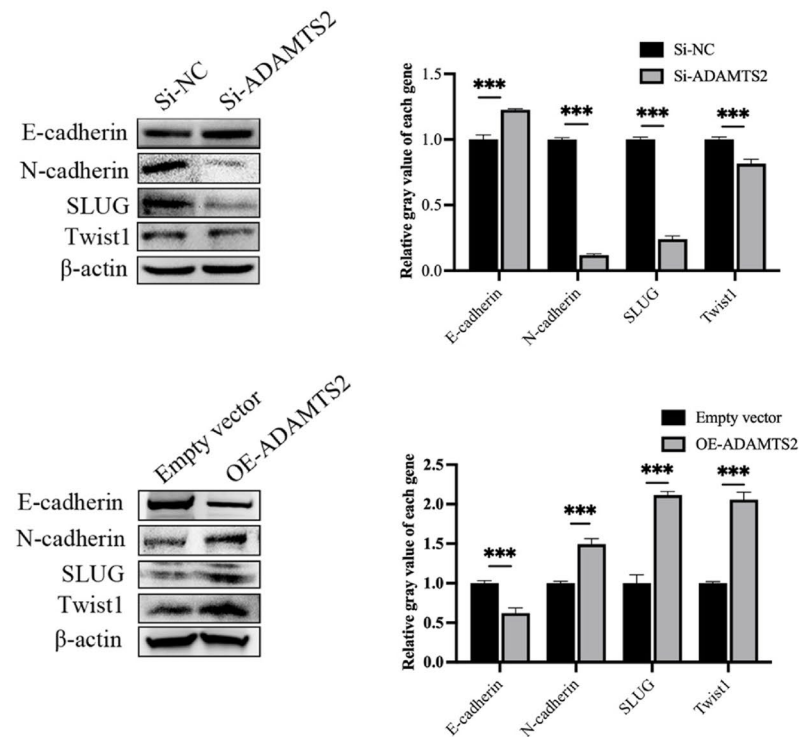


Fig. 10 Effects of ADAMTS2 knockdown and overexpression on the expression of EMT proteins in CAOV3 (A) and OVCAR3 (B) cells. All experiments were performed in triplicate. ** $p < 0.01$, *** $p < 0.001$

CAOV3

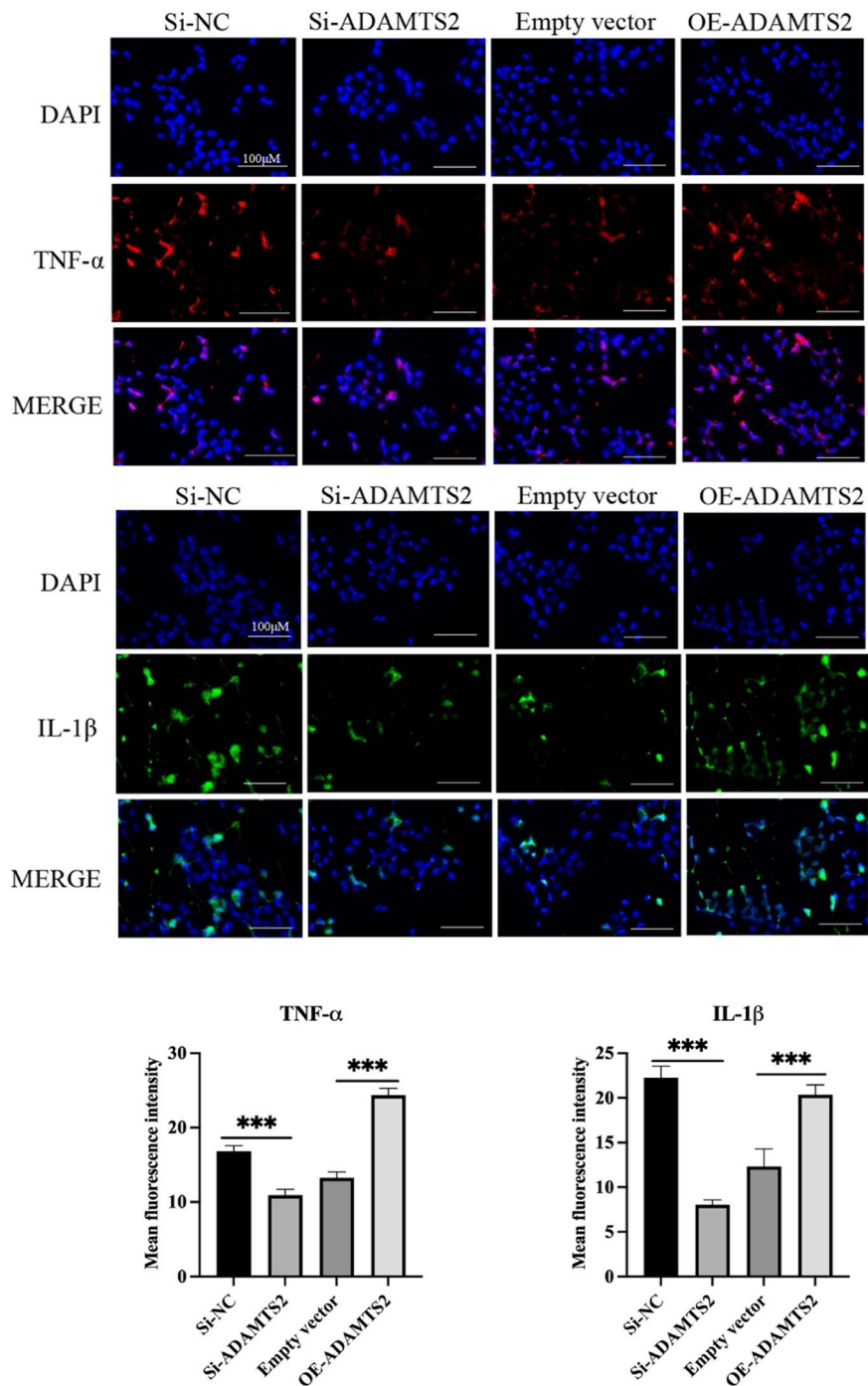


Fig. 11 Effects of ADAMTS2 knockdown and overexpression on the expression of immune-related genes in CAOV3 cells. Representative images of immunofluorescence using specific antibodies. All experiments were performed in triplicate. *** $p < 0.001$

and tumour growth [46, 47]. In addition, FAP expression is associated with tumour stage, metastasis, and survival rate in ovarian cancer patients [48–50]. POSTN, another ECM protein, is an important player in EMT in several cancer types. For example, POSTN promotes EMT and

proliferation in renal cell carcinoma by activating ILK/AKT/mTOR signalling [51]. In hepatoblastoma, POSTN can activate MAPK/ERK signalling and increase the expression of Snail and other EMT-related markers [52]. POSTN can also induce the expression of EMT-related

OVCAR3

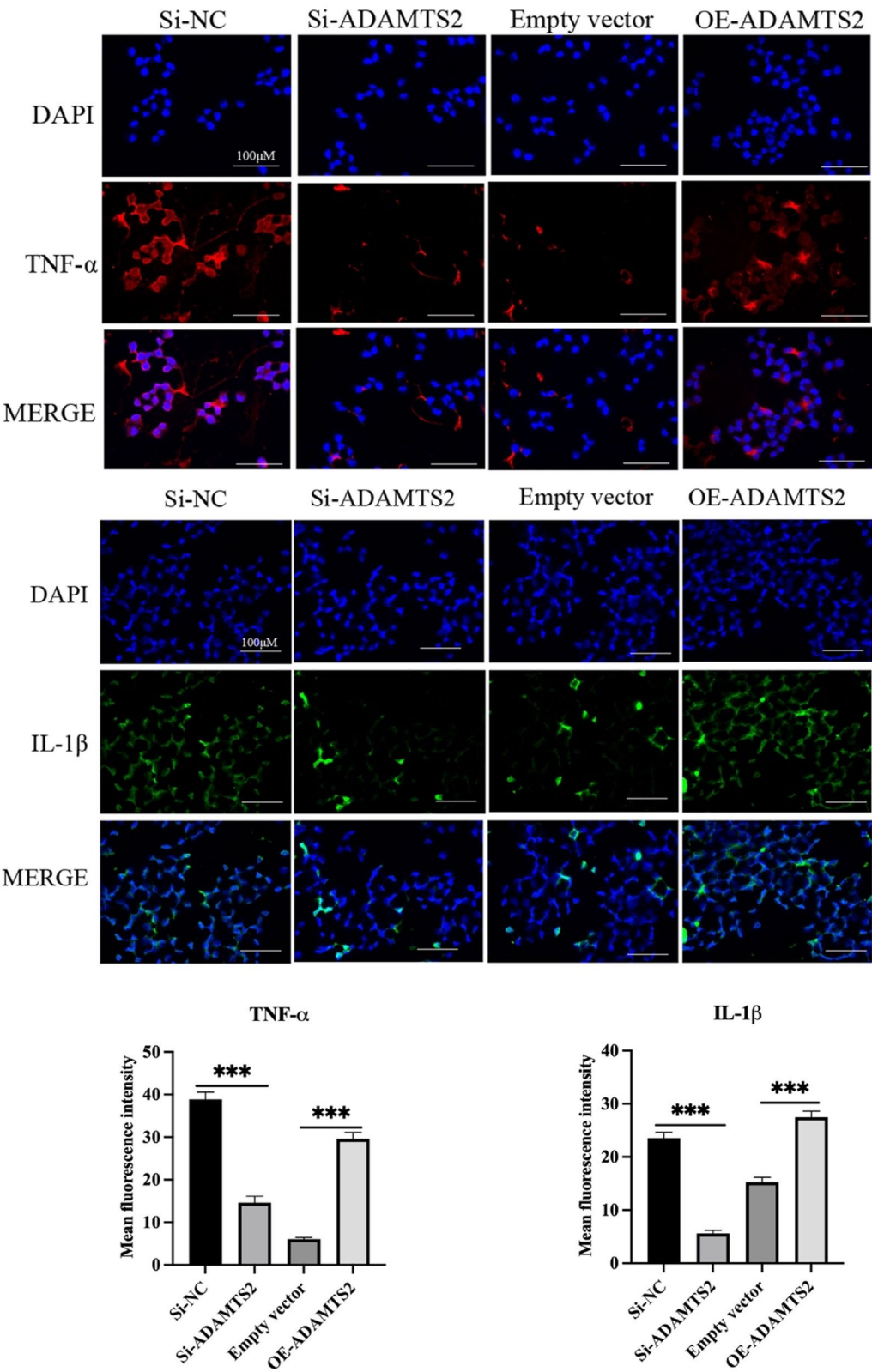


Fig. 12 Effects of ADAMTS2 knockdown or overexpression on the expression of immune-related genes in OVCAR3 cells. Representative images of immunofluorescence using specific antibodies. All experiments were performed in triplicate. ****p* < 0.001

genes, such as *TWIST2*, *MMP9*, and *SNAIL1*, in cholangiocarcinoma [53]. In ovarian cancer, Karlan et al. reported that patients with tumours expressing genes associated with *POSTN/TGFB1* had worse overall survival than those with tumours expressing genes associated with *ESR1/WT1* [54]. In addition, *POSTN* is highly expressed in cancer-associated fibroblasts and binds with integrin $\alpha\beta3$ to promote cell migration and invasion by inducing EMT and activating PI3K/Akt signalling [55]. Considering the good performance of the EMT classifier based on the above genes in predicting the tumour microenvironment and potential therapeutic response, these key signatures may perform EMT-related functions in HGSOC via the tumour microenvironment.

Finally, we focused on the role of the ECM-related protein ADAMTS2 in HGSOC. In vivo and in vitro experiments revealed that ADAMTS2 expression was elevated in HGSOC tissues and promoted the proliferation, migration and invasion of HGSOC cells. In addition, the expression of EMT-related proteins and inflammatory genes was altered after ADAMTS2 knockdown and overexpression, indicating that ADAMTS2 is involved in EMT and plays an important role in EMT–HGSOC tumour microenvironment communication.

There are several limitations to our study. First, to determine how key signatures bridge the EMT process and tumour microenvironment, further studies using in vivo and in vitro models are needed. Second, the performance of the classifier in predicting the therapeutic response of HGSOC patients needs to be verified in clinical practice using a large HGSOC population. Third, the EMT score was derived from transcriptomic data and did not directly assess morphological features such as spindle-like cell shape or epithelial polarity loss. Given the emerging concept of partial EMT, in which cells exhibit both epithelial and mesenchymal traits, transcriptomic changes may not fully reflect morphological transitions. Future studies integrating transcriptomic, histological, and single-cell morphological analyses will be necessary to better characterise the extent of EMT in HGSOC. Additionally, while our findings demonstrate that ADAMTS2 modulates the expression of proinflammatory cytokines such as $\text{TNF-}\alpha$ and $\text{IL-1}\beta$ in HGSOC cells, the underlying signalling pathways remain to be elucidated. Based on previous studies, ADAMTS2, through its role in extracellular matrix remodelling, may activate signalling cascades involving $\text{TGF-}\beta$ or VEGF signalling pathways [56, 57], which are well-established regulators of both EMT and inflammatory responses [58–61]. Future studies are needed to explore these mechanistic links. For example, RNA sequencing should be performed following ADAMTS2 overexpression or knockdown to identify key transcriptional changes and affected pathways. Moreover, rescue experiments by

combining ADAMTS2 modulation with specific pathway inhibitors are needed to determine whether blocking these pathways can reverse ADAMTS2-induced effects. These approaches will help clarify the causal mechanisms by which ADAMTS2 contributes to EMT and inflammation in the tumour microenvironment of HGSOC.

Conclusion

For the first time, our study developed a novel EMT classifier that has the potential to predict the immune status and therapeutic response of HGSOC patients. Moreover, we identified *ADAMTS2* as a novel oncogene that promotes EMT in HGSOC. Our findings revealed the importance of the interplay between EMT and the tumour microenvironment in HGSOC, which enhances our understanding of HGSOC pathogenesis from the perspective of EMT at the molecular level.

Abbreviations

HGSOC	high-grade serous ovarian cancer
EOC	epithelial ovarian carcinoma
EMT	epithelial–mesenchymal transition
CNV	copy number variation
MMP2	matrix metalloproteinase 2
ADAMTS2	ADAM metalloproteinase with thrombospondin type 1 motif 2
FN1	fibronectin 1
THBS2	Thrombospondin 2
C3ORF80	Chromosome 3 open-reading frame 80
FAP	Fibroblast activation protein alpha
POSTN	periostin
GAPDH	Glyceraldehyde-3-phosphate dehydrogenase

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12885-025-14649-0>.

Supplementary Material 1.

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Authors' contributions

YT was a major contributor in analysing and interpreting the data and writing the manuscript. JL and RD also analysed the data, especially on the part regarding clinical phenotypes. YG and YW collected tissue samples and performed the qPCR experiments. LY and JZ analysed the qPCR data and plotted the qPCR results. SK supervised the whole project and revised the manuscript. All authors read and approved the final manuscript.

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The study was conducted in accordance with the Declaration of Helsinki and approved by the Ethics Committee of the Fourth Hospital of Hebei Medical

University. Informed consent was obtained from all subjects involved in the study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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